

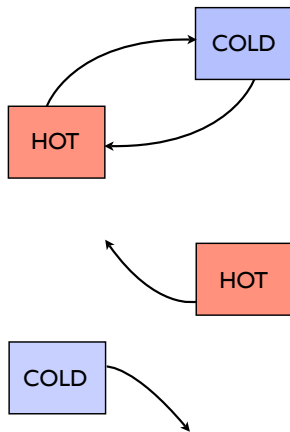
Ocean Mixing

Mixing makes the ocean go round (with caveats).

Sandstrom's Effect

Roughly speaking, if diapycnal mixing is small and if there is no mechanical forcing the heating needs to be at a lower level than the cooling of a fluid in order to maintain a vigorous overturning circulation.

Winds and tides can lead to diapycnal mixing and enable such a buoyancy-driven circulation to occur.



Conclusion: the overturning circulation of the ocean would be noticeably weaker, and so the climate would be different, without mixing.